KAIZEN MATHS

Lesson starter

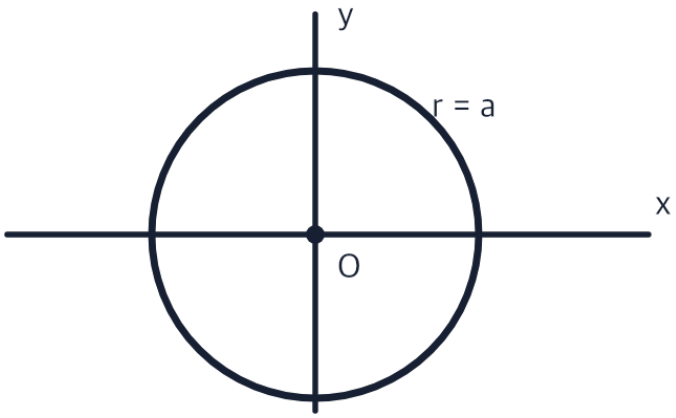
Polar Coordinates and Polar Curves

Polar Curves

Name _____

SA1

Describe each polar curve.



Describe the polar curve r = 7.

Show your working

KAIZEN MATHS

Lesson starter

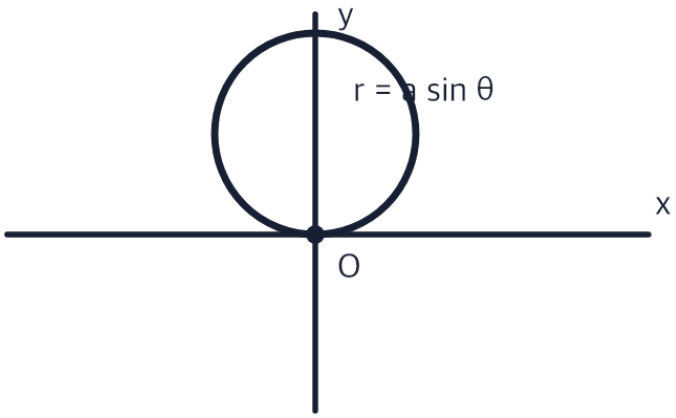
Polar Coordinates and Polar Curves

Polar Curves

Name _____

SB1

Convert each polar circle into Cartesian form and identify its centre and radius.



Convert r = 10 sin θ into Cartesian form and identify the circle.

Show your working

KAIZEN MATHS

Lesson starter

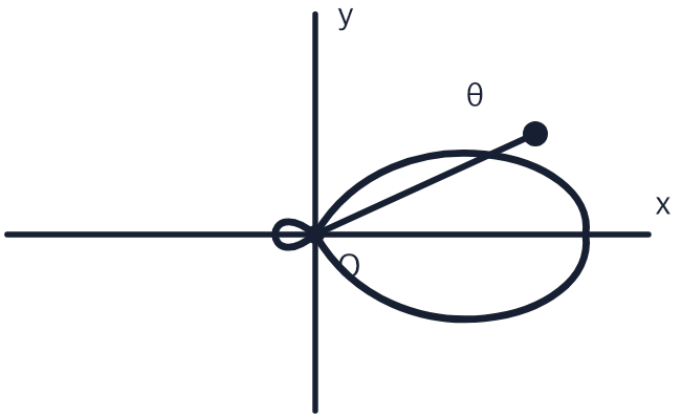
Polar Coordinates and Polar Curves

Polar Curves

Name _____

SC1

Substitute the given angle into the polar equation.



For the curve $r = 3 + 3\cos\theta$, find r when $\theta = \pi/6$.

ALG-294F-L2 · SC

Show your working

KAIZEN MATHS

Lesson starter

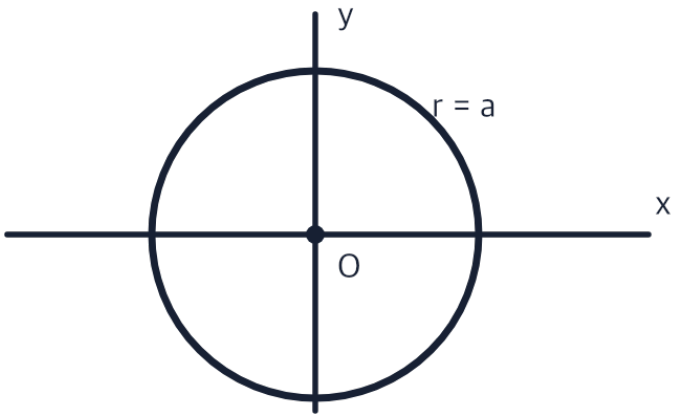
Polar Coordinates and Polar Curves

Polar Curves

Name _____

SD1

Describe each polar curve.



Describe the polar curve r = 6.

KAIZEN MATHS

Exit ticket

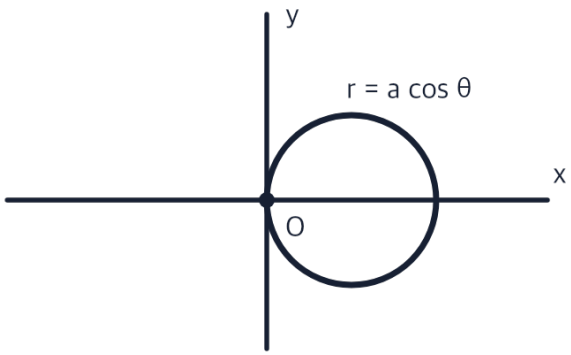
Polar Coordinates and Polar Curves

Polar Curves

Name _____

EA1

Convert each polar circle into Cartesian form and identify its centre and radius.



Convert r = 8 cos θ into Cartesian form and identify the circle.

Show your working

KAIZEN MATHS

Exit ticket

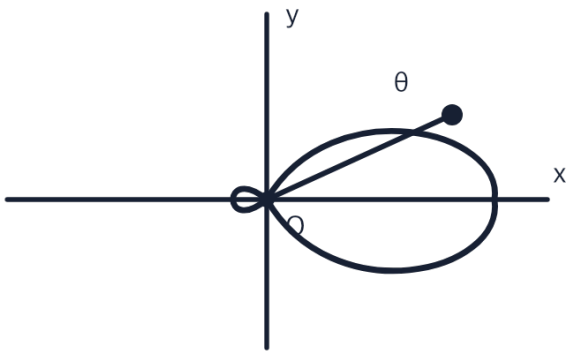
Polar Coordinates and Polar Curves

Polar Curves

Name _____

EB1

Substitute the given angle into the polar equation.



For the curve r = 2 + 2 cos θ, find r when θ = π/4.

Show your working

KAIZEN MATHS

Exit ticket

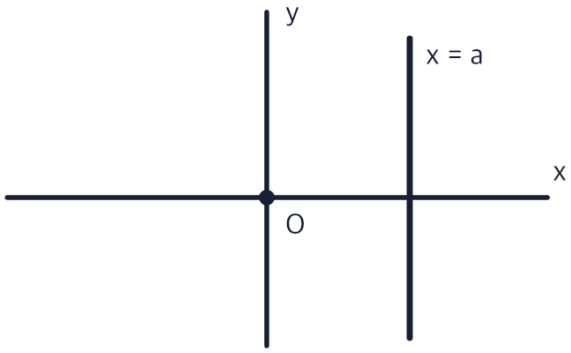
Polar Coordinates and Polar Curves

Polar Curves

Name _____

EC1

Describe each polar curve.



Describe the polar curve r cos θ = 7.

Show your working

KAIZEN MATHS

Exit ticket

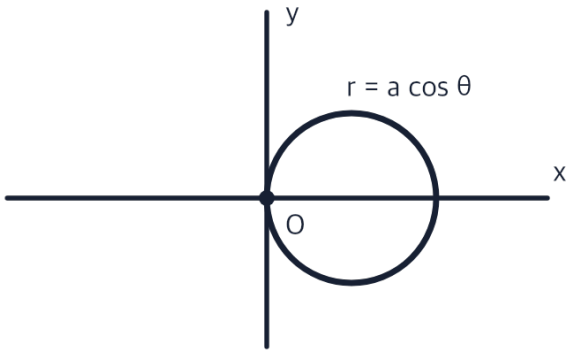
Polar Coordinates and Polar Curves

Polar Curves

Name _____

EA1

Convert each polar circle into Cartesian form and identify its centre and radius.



Convert r = 8 cos θ into Cartesian form and identify the circle.

Show your working

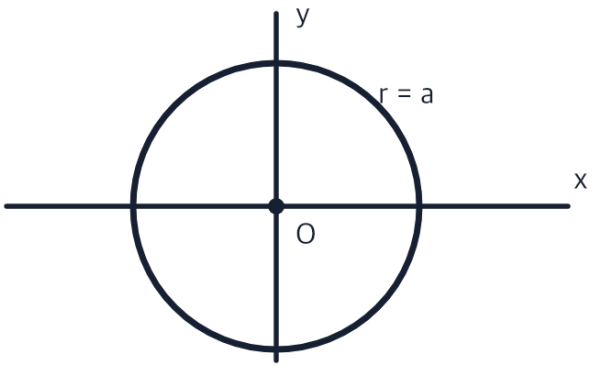
Teacher answers & working

Polar Coordinates and Polar Curves · Polar Curves

ALG-294F-L2 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

SA1

Describe each polar curve.



Describe the polar curve r = 7.

Answer:

A circle centred at the pole with radius 7

Working:

The value of r is the distance from the pole.

The equation r = 7 means every point is 7 units from the pole.

Therefore the curve is a circle centred at the pole with radius 7.

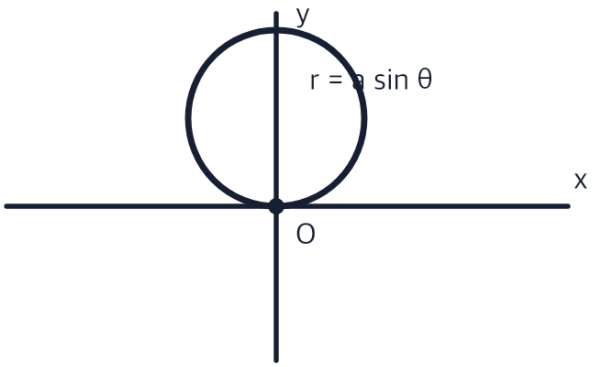
Teacher answers & working

Polar Coordinates and Polar Curves · Polar Curves

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SB1

Convert each polar circle into Cartesian form and identify its centre and radius.



Convert r = 10 sin θ into Cartesian form and identify the circle.

Answer:

$x^2 + y^2 = 10y$; centre (0, 5), radius 5

Working:

Start with r = 10 sin θ.

Multiply both sides by r.

$r^2 = 10r \sin \theta$.

Use $r^2 = x^2 + y^2$, $x = r \cos \theta$, and $y = r \sin \theta$.

So $x^2 + y^2 = 10y$.

$x^2 + y^2 - 10y = 0$.

$x^2 + (y - 5)^2 = 5^2$.

Centre (0, 5), radius 5.

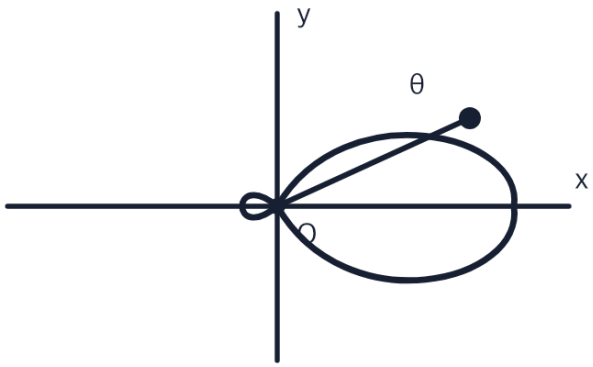
Teacher answers & working

Polar Coordinates and Polar Curves · Polar Curves

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SC1

Substitute the given angle into the polar equation.



For the curve r = 3 + 3 cos θ, find r when θ = π/6.

Answer:

$3 + \frac{3\sqrt{3}}{2}$

Working:

Substitute θ = π/6 into r = 3 + 3 cos θ.

$r = 3 + 3\cos(\pi/6)$.

$\cos(\pi/6) = \frac{\sqrt{3}}{2}$.

Therefore r = 3 + $\frac{3\sqrt{3}}{2}$.

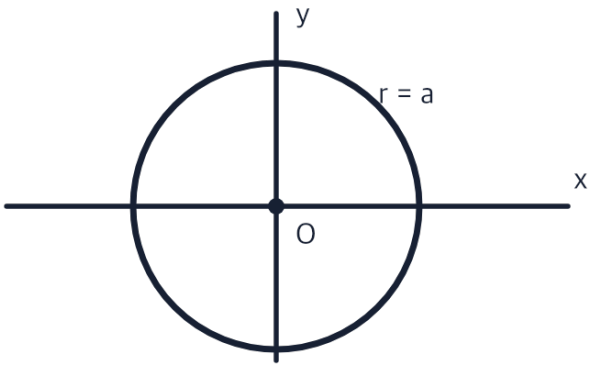
Teacher answers & working

Polar Coordinates and Polar Curves · Polar Curves

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SD1

Describe each polar curve.



Describe the polar curve r = 6.

Answer:

A circle centred at the pole with radius 6

Working:

The value of r is the distance from the pole.

The equation r = 6 means every point is 6 units from the pole.

Therefore the curve is a circle centred at the pole with radius 6.

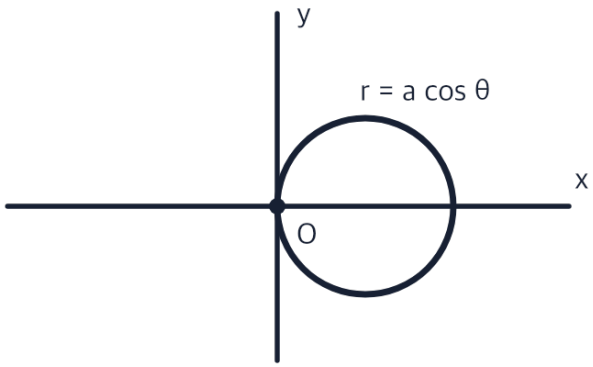
Teacher answers & working

Polar Coordinates and Polar Curves · Polar Curves

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EA1

Convert each polar circle into Cartesian form and identify its centre and radius.



Convert r = 8 cos θ into Cartesian form and identify the circle.

Answer:

$x^2 + y^2 = 8x$; centre (4, 0), radius 4

Working:

Start with r = 8 cos θ.

Multiply both sides by r.

$r^2 = 8r \cos \theta$.

Use $r^2 = x^2 + y^2$, $x = r \cos \theta$, and $y = r \sin \theta$.

So $x^2 + y^2 = 8x$.

$x^2 - 8x + y^2 = 0$.

$(x - 4)^2 + y^2 = 4^2$.

Centre (4, 0), radius 4.

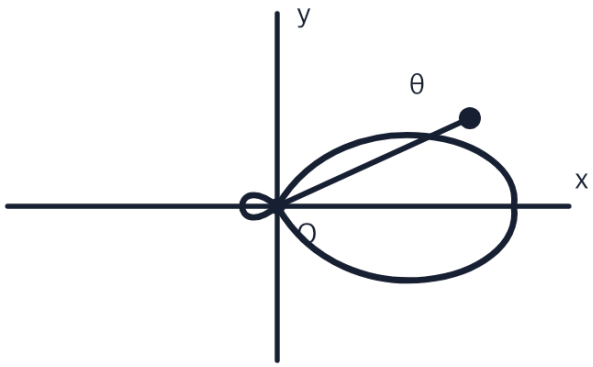
Teacher answers & working

Polar Coordinates and Polar Curves · Polar Curves

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EB1

Substitute the given angle into the polar equation.



For the curve r = 2 + 2 cos θ, find r when θ = π/4.

Answer:

$2 + \sqrt{2}$

Working:

Substitute θ = π/4 into r = 2 + 2 cos θ.

$r = 2 + 2\cos(\pi/4)$.

$\cos(\pi/4) = \frac{\sqrt{2}}{2}$.

Therefore r = 2 + √2.

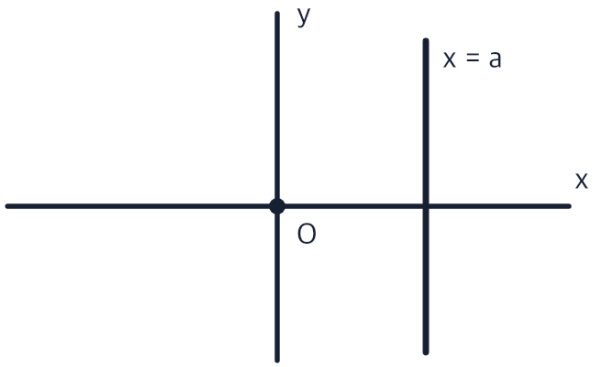
Teacher answers & working

Polar Coordinates and Polar Curves · Polar Curves

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EC1

Describe each polar curve.



Describe the polar curve r cos θ = 7.

Answer:

The vertical line x = 7

Working:

Use $x = r\cos\theta$.

So r cos θ = 7 becomes x = 7.

Therefore the curve is the vertical line x = 7.